

A Metadata-only trajectories

nuScenes metadata
68 logs · 850 scenes

Within-log windows
 $H = 4 \cdot T = 6 \cdot \Delta t = 0.5 \text{ s}$



Input 4×5

Output 6×2

$x \cdot y \cdot \text{speed} \cdot \text{acceleration} \cdot \text{relative yaw}$

Excluded: camera · LiDAR · map · agents · route commands

B Log-exclusive role splits

TRAIN
49 logs · 597 scenes
18,623 windows

VALIDATION A
6 logs · 80 scenes
2,489 windows

VALIDATION B
6 logs · 80 scenes
2,486 windows

HELD-OUT
7 logs · 93 scenes
2,901 windows

Split by log before window extraction

C Development and frozen selection

Five lightweight predictors



MLP · GRU · LSTM
TemporalCNN · Positional Transformer

10 seeds per model

Training split model fitting + normalization

Validation A checkpoint selection

Validation B per-run smoother selection

Frozen pool: 13 candidates



Raw (1) · MA (2) · Exp (3)
Laplacian (3) · SG (2) · QReg (2)

$\epsilon = 0$ · selection is per model–seed run, not per window

Frozen evaluation strategies

Raw



Fixed SG(5,2)



Val-B selected



D Pre-specified primary held-out audit



Controlled primary
held-out evaluation

Accuracy + six-term third-difference decomposition

Boundary + third-difference endpoint + degradation

Frozen history strata
whole-log bootstrap
Internal configurations 1 + 2